

MCQS FOR MIDTERM PREPARATION 'BIF 501'

Created by: **CR. SABA ALWI**

From 1 - 5

Q.1 Drugs target only about maximum proteins.

- a) 250 b) 5000 c) 4000 **d) 500**

Q.2 Scientific name of rice is ?

- a) *Ricea sativam* **b) *Oryza sativa*** c) *Arabidopsis thaliana* d) *Pisum ricea*

Q.3 Gene therapy-used to ?

- a) Treatment b) cure c) prevent disease **d) All**

Q.4 The science which deals with farm animals including cows and sheep is called?

- a) Pharmaceutical science b) Domestic science **c) Veterinary Science** d) Animal husbandry

Q.5 which disease occur in response to an environmental stress..?

- a) Heart disease b) Cystic Fibrosis c) cancers **d) a and c**

Q.6 Control serious pests of cotton, maize and potatoes?

- a) *Chlorobium tepidum* **b) *Bacillus thuringiensis*** c) *Enterococcus faecalis* d) *Deinococcus radiodurans*

Q. 7 Genes transferred into rice to increase levels of

- a) Vitamin A** b) Vitamin B c) Vitamin C d) Vitamin D

Q.8 Nucleotide Sequence Databases store ?

- a) DNA b) cDNA c) EST sequences **d) All**

Q.9 European Molecular biology Lab (EMBL) established in

- a) 1880 b) 1984 **c) 1980** d) 1960

Q.10 is joint venture of NCBI, EMBL and DDBJ?

- a) INSDC** b) NISDC c) INCDS d) INDCS

Q.11 First protein sequence was achieved in 1951 by protein families called..?

- a) Amino acids b) nucleic acids c) lysosomes **d) cytochromes**

Q.12 Atlas of protein sequences (mainly cytochromes) was assembled at in 1960s?

- a) INSDC b) NCBI c) **NBRF** d) PDB

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Q.13 is Collaboration between the SIB and EBI

- a) PIR b) JIPID c) ExPASy d) Swiss-Prot

Q.14 First sequence to be collected was..... Sequence.

- a) DNA b) nucleic acid c) protein d) carbohydrate

Q.15 First attempt to sequence free living organism was launched in late

- a) 1980's b) 1998 c) 1990's d) 1970's

Q.16 paved the way for other genome sequencing projects and also known as 1st published genome.

- a) Human genome b) streptococcus genome c) Bactericidal genome d) *Haemophilus influenzae*

Q.17..... was the first genome database developed in 1989.

- a) AceDB b) NCBI c) DDB d) JIPID

From 6 – 10

Q.18 Gene Expression Omnibus (GEO) has kinds of records.

- a) two b) five c) three d) four

Q.19..... is a Best resource to get microarray and Next Generation Sequencing (RNASeq) data?

- a) GDS b) GEO c) GSE d) GSM

Q.20 PUBMED Comprises of more than citations for biomedical literature from MEDLINE.

- a) 25 millions b) 23 millions c) 24 millions d) 22 millions

Q. is a tool of Uniprot where protein sequence can stored.

- a) slash b) SPIN c) Sequin d) IUB

Q.21 What is the vocabulary used by MEDLINE?

- a) MeSH b) SLasH c) Roman d) Calibri

Q.22 NCBI has an option for submission of Complex sequences and annotations.?

- a) BANKIt b) IUB c) Sequin d) SPIN

Q. NCBI has an option for submission of small data set sequences and annotations.?

- a) BANKIt b) IUB c) Sequin d) SPIN

Q.23 Genbank format is a format

- a) un predictable b) expensive c) standard d) All

Q. Gene Ontology is refers to different anotational terms.

- a) sequence b) structural c) phylogenetic d) functional

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Q.24 PsycINFO searches the literature.

- a) physiological **b) psychological** c) biological d) bioinformatics

Q.25 in Gene Architecture preparation and description of the samples is carried out by

- a) GSM** b) GPL c) GSE d) GDS

From 11 - 15

Q.26 FASTA is the most frequently used format to present ...?

- a) DNA sequences b) Protein sequences c) RNA sequences **d) a and b**

Q.27 Sometimes a might be present in the end of a sequence format.

- a) ! **b) *** c) "" d) ^

Q.28 One sequence starts with a line containing the word..,

- a) FORM b) FILE c) TITLE **d) LOCUS**

Q.29 Genbank, DDBJ and EMBL has Formats.,

- a) Different b) complicated **c) similar** d) unpredictable

Q.30 The Protocol (FTP) is a standard network protocol used to files .

- a) generate b) assemble c) modify **d) transfer**

Q.31 PERL and Python are good for Biological data.

- a) search b) store **c) process** d) All

Q.32 Study of all of a person's genes (the Genome) is called

- a) Genome informatics **b) Genomics** c) gene study d) taxonomy

Q.33 Which is not a Model organisms from following?

- a) worm b) zebra fish c) you and me **d) alligator**

Q.34 The organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane.

- a) vertebrate b) invertebrate c) eukaryotes **d) Prokaryotes**

Q.35 While Staining with dyes the lighted region of Eukaryotic Genome is ..

- a) introns **b) Euchromatin** c) Heterochromatin d) exon

From 16 – 20

Q.36 Self-replicating additional rings in prokaryotes are

- a) Transposons b) Epichromosomal Elements **c) plasmids** d) cpDNA

Q.37 Which organelle has its own DNA?

- a) Mitochondria b) Golgi bodies c) Chloroplasts **d) a and c**

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Q.38 according to Endosymbiont hypothesis Mitochondria and Chloroplast are derived from..,

- a) Photosynthetic bacteria **b) endosymbiotic bacteria** c) parasitic bacteria d) none

Q.39 How many percent of nuclear genes devoted to mitochondrial function?

- a) 5% b) 12% c) 7% **d) 10%**

Q.40 Repeat containing DNA can be separated as satellite DNA on the bases of their.,

- a) Masses **b) densities** c) functionality d) specificity

Q.41 which can move (Jump) from one location to another faster than chromosome replicate?

- a) Transposable Elements (TEs)** b) Microsatellite c) Minisatellite d) Satellite DNA

Q.42 Insertion Sequences (IS elements) usually very small as,

- a) <2 kb to 3 kb b) <1 kb to 3 kb **c) < 1 Kb to 2 Kb)** d) < 3 kb to 5 kb

Q.43 IS elementsDo not code for noticeable

- a) mutation b) genotypic traits **c) phenotypic traits** d) a and b

Q.44 Example ofClass II TEs.,

- a) LTR b) Retro transposons c) LINES **d) Ac-Ds in maize**

Q.45 Processed pseudogenes lack

- a) introns b) exons **c) a and d** d) promoters

Q.46Introns proportionin Human genes and in yeast is

- a) 95% - 329 b) 97% - 392 c) 99% - 235 **d) 95% and 239**

Q.47 proteome is more similar to mammalian proteomes.

- a) Fruit fly** b) worm c) yeast d) bacteriophage

From 21 – 25

Q.48 Unique set of proteins is called ..,

- a) proteomic b) domain **c) core proteome** d) alpha protein

Q.49 All-against-all comparison provides an indication of Families.

- a)Protein/ Protein b) Gene/ Gene c) Protein/Gene **d) Gene/Protein**

Q.50 Between-Proteome comparisons process helps finding

- a) orthologs b) Gene families c) domains**d) ALL**

Q.51 Expressed Sequence Tags (EST) are cDNA copies of cell's Sequences.

- a) tRNA **b) mRNA** c) DNA d) siRNA

Q.52 To search from proteomes to EST databases is frequently used.

- a) tBLASTx b) BLASTn c) tBLAST d) tBLASTn

Q.52 Transfer of gene is relatively slow process in evolution.,

- a) vertical b) Horizontal c) parallel d) linear

Q.53 Gene mapping is determining the between genes on a chromosome.

- a) difference b) location c) relative distances d) b and c

Q.54 Genetic map distances are based on the genetic linkage information measured in (CM).

- a) Centi-meter b) Celsius meter c) Centi-Morgans d) none

Q.55 Arrangement of genes on the chromosomes is called

- a) loci b) Synteny c) Gene Order d) band c

Q.56 Species diverged from common ancestor have chromosomes and gene order

- a) highest b) lowest c) similar d) different

Q. EST is mainly extracted from.....

- a) introns b) exons c) 3' end d) 5' end

From 26 – 30

Q.57 DNA sequencing began in with the development of Maxam-Gilbert's method.

- a) 1975 b) 1980 c) 1970s d) 1990s

Q. among free living organisms 1st of all a was sequenced.

- a) eukaryotic cell b) drosophila c) an algae d) *Haemophilus influenzae*

Q.58 First complete genome of a phage virus MS2 was sequenced by and colleagues.

- a) Maxam-Gilbert b) Walter Fiers c) Sanger d) none

Q.59 was first sequenced among nucleic acids.

- a) DNA b) Protein c) RNA d) amino acid

Q.60 49 different codons in the genetic code specify the sequence of the amino-acids long coat polypeptide.

- a) 192 b) 60 c) 20 d) 129

Q. ----- is an unstable molecule. It is difficult to isolate it in intact form.

- a) amino acid b) DNA c) RNA d) a and b

Q.61 Leroy E. Hood's laboratory invented the first semi-automated DNA sequencer in.,

- a) 1986 b) 1898 c) 1968 d) 1989

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Q. which one of following is a self-replicating molecule?

- a) lipid **b) DNA** c) RNA d) protein

Q. which enzyme helps in unwinding of DNA strand?

- a) DNA polymerase b) RNA polymerase **c) helicase enzyme** d) protease enzyme

Q.62 Which of the following is not a stop codon?

- a) UAA b) UAG **c) UGG** d) UAA

Q.63 Genome sequencing involves recognition of in a Genome and determining their precise order

- a) amino acids b) protein c) gene loci **d) nucleotides**

Q. which one is the major enzyme for recognition of promoter region during transcription?

- a) DNA polymerase **b) RNA polymerase** c) helicase enzyme d) a and b

From 31 – 35

Q.65

From 36 – 40

Q. Restriction enzyme was 1st discovered in

- a) 1998 b) 1990 **c) 1970** d) 1975

Q. in Block Game if “i” and “j” both have even values the result would be

- a) Loss** b) Win c) Zero d) one

Q. We use Restriction Mapping to calculating

- a) number of R.F **b) size of R.F** c) position of R.F d) All

(here R.F means restriction fragments).

Q.74 Brute Force Algorithm this formula $\binom{n}{2} = \frac{n(n-1)}{2}$ is used for calculating?

- a) Largest distance in “L”** b) Outermost points of “X” c) Remaining distance “δ”

Q.75 List of pair wise distances is represented by,

- a) l b) d **c) L** d) Lt

Q. Brute Force Algorithm was introduced in

- a) 1990** b) 1998 c) 1997 d) 1995

Q.76 Biological function of Nucleases and transcription factors Processes are present at ..,

- a) amino acid level **b) RNA level** c) DNA level d) nucleic acid level

Q. R. motives generally present at?

- a) LTR b) inhibitor region **c) promoter region** d) introns

Q.77 which one is a set of regulatory sequence in drosophila genome,

- a) NF-kB sites** b) Nf – Kbsites c) Nf-kbsites d) NF-KBsites

).

Q. Gold Bug Theory helps us to find ----?

- a) Restriction sites b) LTRs **b) DNA motif** d) cDNA

From 41 – 45

Q.82 consensus score which is use to measure the strength of profile is defined to beScore

- a) (s, RNA) **b) (s, DNA)** c) (t, RNA) d) (c, DNA)

Q. Higher the number of s, DNA, the strength of profile will be

- a) lower **b) higher** c) zero d) medium

Q.83 A consensus score of $l.t/4$, is a alignment

- a) impossible b) negligible **c) worst possible** d) a and c

Q. we can calculate the consensus sequence of any conserved region by

- a) Brute force algorithm b) restriction mapping **c) alignment matrices** d) a and b

Q. we can solve the motive finding problem with the help of

- a) alignment matrices **b) median string** c) brute force d) restriction mapping

Q.84 Median string for DNA as the string v that TotalDistance(v, DNA).

- a) equal to b) inversely proportional c) maximizing **d) minimizes**

Q. Different elements present b/w the two equal length DNA are known as.....

- a) motives b) Blebs **c) Heming distance** d) median string

Q.85 The Median String problem is a Minimization problem and Motif Finding problem a Maximization problem all are Computationally...,

- a) Opposite **b) equal** c) unequal d) different

From 46 – 50

Q. The basic term that an algorithm can use

- a) Array b) variable c) Arguments **d) All**

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Q. 90 Consider all (k^L) L -mers in a k -letter alphabet For Motif Finding problem, $k = \dots\dots\dots$

- a) $n-l+1$, b) $n-l+2$ c) $n-l+3$ d) $n-l+4$

Q. In Max (1 – 80) the value of (1 – 80) is called an

- a) Array b) Argument c) variable d) algorithm

Q.91 Algorithm for Search Trees for searching NEXTLEAF(a, L, k), here the value of ' k ' represents--,

- a) constant length b) number of strings c) number of element d) length of leaf

Q.92 Consider all k^L L -mers in a k -letter alphabet For Median String problem while finding DNA seq, $k = s$

- a) 3 b) 1 c) 4. d) 6

Q.93 In Median String Problem to find DNA seq , $L = \dots\dots\dots$ and $k = 4$

- a) k b) l c) n d) 1

Q.94 In Next Vertex Algorithm when $i < L$, NEXTVERTEX moves $\dots\dots\dots$ to the next lower level and explores that subtree.

- a) left b) right c) up d) down

Q.95 To carry out the steps of an algorithm humans are generally..,

- a) fast b) slow c) quick d) a and c

Q.96 Algorithm must be rephrased in.....,

- a) simple language b) statistically c) programming language d) alphabetically

Q.98 language often used to describe algorithm.

- a) java script b) pascal c) alphabetical d) Pseudocode

Q.99 Complex operations are grouped together into mini-algorithms called ...,

- a) miniatures b) subgroups c) groups d) matrix

Q.100 An of n elements is an ordered collection of n variables.

- a) variable b) argument c) array d) matrix

From 51 – 55

Q.101 In ByPass Algorithm scores at internal vertices represent the in the subtree rooted at that vertex.

- a) maximum score b) minimum score c) total score d) average score

Q.102 In Brute Force Algorithm A search tree for the Median String problem. Each branching point can give rise to only children.

- a) two b) five c) four d) three

Q.103 Q. Branch and bound algorithm is an Algorithm.

- a) time consuming **b) time saving** c) slow d) a and c

Q. In Branch and bound algorithm the (t-i)-l represents the

- a) best score b) known score **c) unknown score** d) maximum score

Q. which algorithm shows the ability of Bypass Algorithm?

- a) Brute force b) motif finding c) simple motives search **d) Next vertex**

Q. Median search problem can be used as an alternative to find out

- a) DNA seq b) Nucleotide seq **c) Motives** d) RNA seq

Q. Name the two algorithms which we merge during finding motives? No option

Ans: 1. Brute Force motives search

2. Next leaf algorithm

From 56 – 60 o

Q.108 Genome rearrangement results in a change of gene ..,

- a) Numbering b) size c) length **d) ordering**

Q.109 For analysis of human and mouse genomes genomic rearrangements required.

- a) 200 b) 356 **c) 250** d) 125

Q.110 is to solve the Motif Finding problem.

- a) greedy algorithm b) By Pass algorithm c) next vertex algorithm **d) Brute force algorithm**

Q.111 GREEDY MOTIF SEARCH scans each DNA sequence only for

- a) 3 times b) 2 times c) 4 times **d) 1 time**

Q.112 cancer-causing -sis oncogene matched a normal gene involved in growth and development called growth factor.

- a) RBCs-derived b) WBCs-derived **c) platelet-derived** d) serum-derived

Q.113 cystic fibrosis produce abnormally thick mucus that clogs the ..,

- a) trachea b) blood vessels c) heart walls **d) lungs**

Q.114 There is a 25% chance that the will have cystic fibrosis.

- a) adult b) old age **c) child** d) women

Q.115 In 1 million nucleotides are discovered on the chromosome 7.

- a) 1898 **b) 1989** c) 1998 d) 1889

Q.116 cystic fibrosis involves sweat secretions with abnormally high content.

a) phosphate b) iron c) calcium d) sodium

Q.117 Manhattan Tourist Problem is to find a..... path in a weighted grid.

a) median b) shortest c) longest d) closest

Q.118 Hamming distance is not typically used to compare sequences.

a) RNA b) DNA c) protein d) b and c

Q.119 Mutation in DNA-evolutionary process: DNA replication- substitutions, insertions, and deletions of nucleotides, leads to DNA texts.

a) "deleted" b) "edited" c) "submitted" d) "substituted"

From 61 – 65

Q.120 Edit distance between two strings as the of editing operations needed to transform one string into another,

a) next number b) previous number c) minimum number d) maximum number

Q.121 In an Alignment Columns that contain the same letter in both rows are called

a) indels b) insertion c) mismatches d) matches

Q.122 The columns of the alignment containing one space are called

a) mismatches b) match c) indels d) score

Q.123 The number of matches plus the number of mismatches plus the number of indels is equal to the of the alignment matrix.

a) total number b) length c) number of strings d) a and b

Q.124 The main difference between the Manhattan and Edit Graph is that we can move along the diagonals in the ...

a) Manhattan b) Edit Graph c) none

Q. Where there is a missing number in the string "v" into the edit graph, it will be represented by

a) Diagonal arrow b) horizontal arrow. c) vertical arrow d) double arrow

Q.125 In edit graph column as a if both letters are the same

a) zero b) one number c) positive number d) negative number

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Q.126 The score for the is the sum of the individual column scores.

- a) positive number b) negative number c) indels d) whole alignment

Q.127 Longest Common Subsequence (LCS) problem is a form of sequence similarity analysis that is..

- a) complicated b) expensive c) simplest d) toughest

Q.128 In LCS we eliminate the operation of substitution and allow only

- a) insertion b) deletion c) modification d) a and b

From 66 – 70

Q.129 Random mutagenesis-change sequence

- a) DNA b) RNA c) nucleic acid d) amino acid

Q.130 Scoring matrices for are complicated – *pointed accepted mutation* (PAM) and *block substitution* (BLOSUM),

- a) DNA sequence b) RNA sequence c) protein sequence d) nucleic acid sequence

Q. In Recurrence of LCS problem $S_{i-1,j}$ represent that is not present in LCS.

- a) W_i b) V_i c) V_j d) W_j

Q.131 The best scoring matrix to compare two proteins depends on similarity of comparing ..

1. a) proteins b) amino acids c) nucleic acid d) organisms

Q.132 If proteins are % identical then "obvious" alignments are constructed that are used to compute scoring matrix δ .

- a) 70 % b) 90% c) 80% d) 100%

Q.134 To transfer "V" string into "W" string we need indels in

- a) W string b) V string c) both d) none

Q. 135 In LCS problem the letter 'k' is equal to When we are dealing with protein sequence.

- a) 4 b) 10 c) 12 d) 20

Q.136 In Global sequence alignment we assign 'u' to the

- a) insertion b) deletion **c) mismatches** d) matches

From 71 – 75

Q.141 *Homeobox* genes, regulate development- variety of species.

- a) structural b) organ **c) embryonic** d) tissue

Q.142 To find conserved area in *Homeobox* genes ignore areas-little.....

- a) differences b) distances c) variability **d) similarity**

Q.143 proposed modification of the global sequence alignment dynamic programming algorithm-

- a) Temple Smith b) Micheal Waterman **c) a and b** d) none

Q.144 Which type of problem extend over entire length?

- a) local alignment **b) global alignment** c) sequence alignment d) none

Q. Sub domain or conserved domain within a gene is often Of the size of the gene.

- a) 2 third **b) 1 third** c) half d) equal

Q.145 In Progressive Multiple Alignment Many popular iterative algorithms done by some tools also including

- a) MS b) Expasy c) FASTA d) **CLUSTAL**

Q.146 In *Multiple Longest Common Subsequence* problem, the score of a column is set to if all the characters in the column are the same.

- a) 0 b) 10 **c) 1** d) 2

Q.147 Sydney Brenner and Francis Crick Introduce deletions in,

- a) DNA** b) protein c) RNA d) amino acids

Q.148 Deleting three consecutive nucleotides results in change in the protein .

- a) total change b) major change **c) minor changed** d) no change

Q.149 Charles Yanofsky proved that a gene and its protein product are...,

- a) antagonistic **b) collinear** c) opposite d) a and b

Q.150 Human genome is and complex than bacterial genome.

- a) small b) mini **c) larger** d) a and b

Q.151 Salamander genome is times larger than the human genome.

- a) five b) eight c) two **d) ten**

Q. Only concern specific part of DNA not whole length..?

- a) LAP** b) GAPGDN

From 76 – 80

Q.152 In Gene Prediction the jump is inconsistent from species to

- a) organism b) genus c) class **d) species.**

Q.153 The gene in insect genome-in worm.

- a) similar **b) different** c) found d) a and c

Q.154 The four continuous segments of Hexon mRNA, exons in the adenovirus genome are separated by three “junk” fragments called.....,

- a) neutron b) duplex **c) introns** d) genome

Q.155 After mRNA-DNA hybridization how many loop structures form?

- a) five b) seven c) two **d) three**

Q.156 In Gene Prediction mRNA is built from four separate fragments of the adenovirus genome.

- a) octane b) butane **c) hexon** d) intron

Q.157 Gene prediction algorithms for tend to be somewhat simpler.

- a) eukaryotes **b) prokaryotes** c) higher animals d) a and c

Q.158 Dinucleotides AG and GT on the left- and right-hand sides of an exon are.....

- a) inter changeable b) least conserved **c) highly conserved** d) non conserved

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Q.159 The quite weak-matches that are frequently found in the genome at non splice sites are ..,

- a) hexon b) intron c) exon d) Profiles

Q.160 99% of have human analogs.

- a) drosophila genes b) chimpanzee genes c) mouse genes d) prokaryotic genes

Q.161 ORFs within a single genomic sequence may overlap since there arepossible "reading frames":

- a) three b) two c) four d) six

Q.162 gene prediction algorithms based on selecting significantly long ORFs may fail to detect ..,

- a) short amino acids b) short genes c) long genes d) long amino acids

Q.163 We can enter the frequency of occurrence of each codon within a given sequence into a-element *codon usage array*.

- a) 56 b) 34 c) 20 d) 64

Q.164 Gene prediction in bacterial genomes-several conserved sequence motifs often found in the regions around the start of...,

- a) splicing b) translation c) transcription d) prediction

From 81 – 85

Q.165 a weakly conserved sequence of exist at the boundary of an exon and an intron (donor splice site).

- a) 3 amino acids b) eight nucleotides c) six nucleotides d) seven nucleotides

Q.166 GENSCAN combines and splicing signal predictions into a single framework.

- a) Conserved region b) non coding region c) splicing region d) coding region

Q.167) The accuracy of GENSCAN decreases for genes with many....,

- a) long introns b) short introns c) short exons d) long exons

Q.168) In Gene Prediction Brute force approach find all,

- a) local similarities b) local differences c) global similarities d) global differences

Q.169) An ORF is a sequence of that starts with start codon "ATG" (not always) and ends with any of the three termination codons (TAA, TAG, TGA)

- a) protein b) cDNA c) DNA d) RNA

Q.170) Generally ORFs are preferred.

- a) shorter b) longer c) medium d) none

Q.171) ORF provide an important evidence in,

- a) sequencing b) splicing c) a and b d) gene finding

Q. How many percentage of junk DNA

Q.172) Depending on the start point, there are possible ways of translating any nucleotide sequence into amino acid sequence according to the genetic code.

- a) 20 b) 9 c) 12 d) 6

Q.173) Six possible ways of translating any nucleotide sequence are found, Three on forward strand and three on

- a) mRNA. b) ORF c) complementary Strand d) genetic code

Q.174) While ORF Finding Long ORF may be a gene Expected, i.e. codons before we see a stop codon

- a) $64/3 \sim 21$ b) $65/4 \sim 21$ c) $62/3 \sim 22$ d) $64/4 \sim 21$

Q.175) An ORF is more 'reliable' if it has 'likely',

- a) genes b) amino acids c) length d) codons

Q. How many % of human genome coded into protein.?

- a) 50% b) 15% c) 10% d) 3%

Q.176) However average vertebrate exon length is nucleotides.

- a) 120 b) 130 c) 140 d) 150

From 86 – 90

Q.178) Translation starts with ATG that codes for in a polypeptide

- a) proline b) cysteine **c) methionine** d) thiamine

**Q.179) Hexamer frequencies in coding versus regions may provide important insights
Frequency of X(A,G,C,T) at position i.**

- a) mutated **b) non-coding** c) nucleotide d) coding

**Q.180) In translation start sequence mathematical model we get a score, If the score is larger than
....., then it is a "translation start.**

- a) one b) twenty c) two **d) zero**

Q.181) TSS can be predicted while using the of neighboring nucleotides.

- a) length b) distance **c) frequency** d) number

Q.182) Like TSS, the flanks of show "biased" distributions of nucleotides in certain positions.

- a) motives **b) splice junctions** c) ORFs d) amino acids

Q.183) These biased distributions of can be used for prediction of splice junctions

- a) amino acids b) DNA c) RNA **d) nucleotides**

Q.184) can be predicted within ORF from the information gathered about splice junctions.

- a) gene locus b) intron **c) Exons** d) b and c

**Q.185) Much of the data processing in bioinformatics involves searching and recognizing certain
patterns within ...**

- a) DNA b) RNA c) protein sequences **d) All**

**Q.186) After a genome is assembled, genome are performed to identify gene and other features
in a genome.**

- a) sequencing b) prediction **c) annotations** d) splicing

Q.187) A pattern (keyword) is an ordered sequence of ,,

- a) alphabets b) numbers c) genes **d) symbols**

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Q.188) Symbols of the pattern and the searched text are chosen from a predetermined finite set, called an

- a) alphabets b) numbers c) genes d) symbols

From 91 – 95

Q.190) Exact Pattern Matching Algorithms is.,

- a) Naïve Brute Force algorithm b) Boyer-Moore algorithm c) Knuth Morris Pratt algorithm d) All

Q.191) Example of Approximate Pattern Matching Algorithms is

- a) Dynamic programming approach b) Automata approach c) Knuth Morris algorithm d) a and b

Q.192) Position Also known as position specific scoring matrices (PSSM)

- a) mass matrices b) Weight Matrices c) number matrices d) distance matrices

Q.193) A matrix representing the frequencies of residues observed for a position in multiple alignment is

- a) Position Weight Matrices b) Exact searching matrices c) Suffix trees d) Naïve Brute Force

Q.194) A compressed tree containing all the suffixes and allows many problems on strings to be solved quickly is called ..,

- a) rooted tree b) unrooted tree c) Suffix Trees d) symbolic tree

Q.195) A linear time algorithm for string matching which does not involve backtracking is called ,,

- a) Naïve Brute Force algorithm b) Boyer-Moore algorithm c) Knuth Morris Pratt algorithm d) All

Q.196) A linear time algorithm for string matching avoid useless shifts of the,

- a) symbols b) pattern c) components d) scores

Q.197) Knuth-Morris-Pratt Algorithm is used to compute the prefix function values and is to compare the pattern to the text.

- a) $O(N)$, $O(M)$ b) $O(p)$, $O(t)$ c) $O(m)$, $O(n)$ d) $O(t)$, $O(p)$

Q.198) The running time of the KMP algorithm is optimal ($O(m + n)$), which is very

- a) fast b) slow c) moderate d) none

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Q.199)Knuth-Morris-Pratt algorithm needs to move backwards that makes the algorithm good for processing very large files.

- a) always b) sometimes c) never d) all times

Q.200)Knuth-Morris-Pratt Algorithm is a fast linear time algorithm for matching.

- a) nucleotides b) motives c) amino acids d) string

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